

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250283230

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

SUMI; Shintaro

METHOD FOR PRODUCING ELECTROLYSIS CELL

Abstract

A method for producing an electrolysis cell includes a joining step of joining a frame portion of a protective sheet member provided between a membrane electrode assembly and a fluid-supply-side current collector to a portion of the membrane electrode assembly on the outer side of the covered portion where an electrolyte membrane is covered with an electrode catalyst layer to form a joint, and a joined body stacking step of stacking the membrane electrode assembly and the protective sheet member joined together on the fluid-supply-side current collector with the protective sheet member facing the fluid-supply-side current collector.

Inventors: SUMI; Shintaro (WAKO-SHI, JP)

Applicant: HONDA MOTOR CO., LTD. (TOKYO, JP)

Family ID: 96948646

Appl. No.: 19/052923

Filed: February 13, 2025

Foreign Application Priority Data

JP 2024-037066

Mar. 11, 2024

Publication Classification

Int. Cl.: C25B9/23 (20210101); C25B15/08 (20060101)

U.S. Cl.:

CPC C25B9/23 (20210101); C25B15/08 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-037066 filed on Mar. 11, 2024, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present disclosure relates to a method for producing an electrolysis cell.

Description of the Related Art

[0003] There is an electrolysis stack including a stack body in which a plurality of electrolysis cells are stacked. Examples of the electrolysis cell include a water electrolysis cell that electrolyzes water and an electrochemical hydrogen compressor cell that electrolyzes hydrogen (hydrogen gas). The electrolysis stack including a stack body in which a plurality of water electrolysis cells are stacked is sometimes referred to as a water electrolysis device. The electrolysis stack including a stack body in which a plurality of electrochemical hydrogen compressor cells are stacked is sometimes referred to as an electrochemical hydrogen compressor (EHC).

[0004] JP 2019-157212 A discloses a water electrolysis device including water electrolysis cells. In this water electrolysis device, a plurality of water electrolysis cells are stacked. The water electrolysis cell includes a membrane electrode assembly, a pair of current collectors provided on both sides of the membrane electrode assembly, and a protection sheet member provided between one of the pair of current collectors and the membrane electrode assembly.

SUMMARY OF THE INVENTION

[0005] The electrolysis cell is produced by stacking the membrane electrode assembly, the pair of current collectors, and the protective sheet member in a predetermined order. It is desired to reduce the risk of misalignment in producing the electrolysis cell.

[0006] The present invention has the object of solving the aforementioned problem.

[0007] According to an aspect of the present disclosure, there is provided a method for producing an electrolysis cell including: a membrane electrode assembly including an electrolyte membrane with a through hole and a pair of electrode catalyst layers provided respectively on both surfaces of the electrolyte membrane and covering the electrolyte membrane in a range extending between a portion surrounding a periphery of the through hole to a marginal edge portion of an outer circumferential edge; a pair of current collectors provided on both sides of the membrane electrode assembly; and a protective sheet member disposed between one of the pair of current collectors and the membrane electrode assembly, wherein the one of the pair of current collectors is a fluid-supply-side current collector to which a fluid to be used for electrolysis is supplied, and the protective sheet member includes an inner portion facing the electrolyte membrane in the range covered with the electrode catalyst layers and a frame portion surrounding the inner portion, the method for producing the electrolysis cell including a joining step of forming a joint by joining the frame portion to a portion of the membrane electrode assembly at a position outward of the range covered with the catalyst layers; and a joined body stacking step of stacking on the fluid-supply-side current collector the membrane electrode assembly and the protective sheet member that have been joined, in a manner that the protective sheet member faces the fluid-supply-side current collector.

[0008] According to the aspect of the present disclosure, even if there is no dedicated positioning member for the protective sheet member, the misalignment of the protective sheet member can be prevented. As a result, it becomes possible to reduce the risk of misalignment in producing the electrolysis cell.

[0009] The above and other objects, features and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying

drawings in which a preferred embodiment of the present invention is shown by way of illustrative example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a perspective view of an electrolysis stack according to an embodiment;
[0011] FIG. 2 is an exploded perspective view of an electrolysis cell forming the electrolysis stack;
[0012] FIG. 3 is a cross-sectional view taken along line III-III of FIG. 2;
[0013] FIG. 4 is a flowchart illustrating a procedure of a method for producing the electrolysis cell;
[0014] FIG. 5 is a diagram showing a joined body joined at a joint;
[0015] FIG. 6 is a diagram showing another joined body joined at a joint;
[0016] FIG. 7 is a diagram showing a state after a first arrangement step; and
[0017] FIG. 8 is a diagram showing a state in which the joined body shown in FIG. 5 is stacked.

DETAILED DESCRIPTION OF THE INVENTION

[0018] As shown in FIG. 1, an electrolysis stack **10** according to the present embodiment includes a stack body **14** in which a plurality of electrolysis cells **12** are stacked in a vertical direction (arrow A direction) or a horizontal direction (arrow B direction). The electrolysis cell **12** may be a water electrolysis cell. Alternatively, the electrolysis cell **12** may be an electrochemical hydrogen compressor cell.

[0019] The electrolysis stack **10** includes a plurality of electrolysis cells **12**, a pair of terminal plates **16a**, **16b**, a pair of insulating plates **18a**, **18b**, and a pair of end plates **20a**, **20b**. The terminal plate **16a**, the insulating plate **18a**, and the end plate **20a** are sequentially arranged in this order upward at one end (upper end) of the stack body **14** in the stacking direction. Similarly, a terminal plate **16b**, an insulating plate **18b**, and an end plate **20b** are sequentially arranged in this order downward at the other end (lower end) of the stack body **14** in the stacking direction.

[0020] The electrolysis stack **10** is fastened in the stacking direction. For example, the electrolysis stack **10** is held in a state where the end plates **20a**, **20b** are integrally fastened by a pressing mechanism such as a plurality of tie rods **22** extending in the arrow A direction. Moreover, the electrolysis stack **10** may also be retained together integrally by a box-like casing (not shown) formed of plates serving as the end plates **20a**, **20b**. Further, although the electrolysis stack **10** has a substantially cylindrical columnar shape as a whole, the overall shape thereof can be set to any of various shapes, such as a cubic shape.

[0021] Terminal portions **24a**, **24b** are provided in an outwardly projecting manner on side portions of the terminal plates **16a**, **16b**. The terminal portions **24a**, **24b** are electrically connected via wirings **26a**, **26b** to an electrolytic power source **28**.

[0022] As shown in FIGS. 2 and 3, each of the electrolysis cells **12** includes a substantially disc-shaped membrane electrode assembly **30**, and a first separator **32** and a second separator **34** which sandwich the membrane electrode assembly **30** and the like therebetween. Between the first separator **32** and the second separator **34**, a resin frame member **36** is disposed in surrounding relation to the membrane electrode assembly **30**.

[0023] The resin frame member **36** is of a substantially ring-like shape, and seal members **37a**, **37b** (see FIG. 3) are provided respectively on both surfaces of the resin frame member **36**. One end in a diametrical direction (the arrow B direction) of the resin frame member **36** is provided with a fluid inlet **38a** extending in the stacking direction (the arrow A direction). The fluid inlet **38a** is formed to introduce a fluid used for electrolysis. The fluid inlets **38a** of the plurality of stacked electrolysis cells **12** communicate with each other. The other end in the diametrical direction (the arrow B direction) of the resin frame member **36** is provided with a fluid outlet **38b** extending in the stacking direction (the arrow A direction). The fluid outlet **38b** is formed to discharge a mixed fluid

containing an unreacted fluid that has not been electrolyzed. The fluid outlets **38b** of the plurality of stacked electrolysis cells **12** communicate with each other.

[0024] As shown in FIG. 1, a fluid introduction opening **39a** is provided on a side of the resin frame member **36** of the lowermost electrolysis cell **12** positioned at one end (lower end) in the stacking direction (arrow A direction) among the plurality of electrolysis cells **12**. The fluid introduction opening **39a** is connected to the fluid inlets **38a** (see FIGS. 2 and 3). A fluid lead-out opening **39b** is provided on a side of the resin frame member **36** of the uppermost electrolysis cell **12** positioned at the other end (upper end) in the stacking direction (arrow A direction) among the plurality of electrolysis cells **12**. The fluid lead-out opening **39b** is connected to the fluid outlets **38b** (see FIGS. 2 and 3).

[0025] As shown in FIGS. 2 and 3, a high pressure fluid discharging hole **38c** penetrates through the diametrical central portion of the electrolysis cells **12** in the stacking direction. The high pressure fluid discharging hole **38c** is formed to increase the pressure of a gas generated by electrolysis of the fluid used for electrolysis and discharge the gas. The high pressure fluid discharging holes **38c** of the plurality of stacked electrolysis cells **12** communicate with each other. The gas flowing to the high pressure fluid discharging hole **38c** is discharged in a state of being pressurized to, for example, 1 MPa to 80 MPa.

[0026] The first separator **32** and the second separator **34** are substantially disc-shaped and are made, for example, of carbon members or the like. Apart therefrom, the first separator **32** and the second separator **34** may be formed by press forming steel plates, stainless steel plates, titanium plates, aluminum plates, steel plates subjected to a plating process, or alternatively, metal plates subjected to an anti-corrosive surface treatment on the metal surfaces thereof. Alternatively, the first separator **32** and the second separator **34** may also be formed by applying an anti-corrosive surface treatment after having carried out a cutting process.

[0027] A substantially ring shaped electrolyte membrane **40** made of a solid polymer, a first electrode catalyst layer **42a**, a second electrode catalyst layer **44a** form the membrane electrode assembly **30**. The membrane electrode assembly **30** is disposed between the first current collector **42** and the second current collector **44**.

[0028] The electrolyte membrane **40** is an ion exchange membrane. The first electrode catalyst layer **42a** is an electrode catalyst layer (fluid-supply-side electrode catalyst layer) to which a fluid used for electrolysis is supplied. The first current collector **42** is a current collector (fluid-supply-side current collector) to which a fluid used for electrolysis is supplied. The second electrode catalyst layer **44a** is an electrode catalyst layer on the opposite side from the first electrode catalyst layer **42a**. The second current collector **44** is a current collector on the opposite side from the first current collector **42**. The first electrode catalyst layer **42a** may be simply referred to as the electrode catalyst layer **42a**. The same applies to the second electrode catalyst layer **44a**. The first current collector **42** may be simply referred to as a current collector **42**. The same applies to the second current collector **44**.

[0029] When the electrolysis cell **12** is a water electrolysis cell, the electrolyte membrane **40** may be an anion exchange membrane or a proton exchange membrane. In the case where the electrolyte membrane **40** is an anion exchange membrane, the fluid used for electrolysis is alkaline water. In the case where the electrolyte membrane **40** is an anion exchange membrane, the electrode catalyst layer **42a** and the current collector **42** function as the anode, and the electrode catalyst layer **44a** and the current collector **44** function as the cathode, the gas flowing through the high pressure fluid discharging hole **38c** (see FIG. 1) is hydrogen generated by electrolysis. In this case, the mixed fluid discharged from the fluid lead-out opening **39b** (see FIG. 1) contains unreacted water that has not been electrolyzed and oxygen that has been generated by electrolysis. On the other hand, in the case where the electrolyte membrane **40** is a proton exchange membrane, the electrode catalyst layer **42a** and the current collector **42** function as the cathode, and the electrode catalyst layer **44a** and the current collector **44** function as the anode, the gas flowing through the high pressure fluid

discharging hole **38c** (see FIG. 1) is oxygen that is generated by electrolysis. In this case, the mixed fluid discharged from the fluid lead-out opening **39b** (see FIG. 1) contains unreacted water that has not been electrolyzed and hydrogen generated by electrolysis.

[0030] In the case where the electrolysis cell **12** is a water electrolysis cell and the electrolyte membrane **40** is a proton exchange membrane, the fluid used for electrolysis is water containing a predetermined amount or less of impurities (for example, pure water). In the case where the electrolyte membrane **40** is a proton exchange membrane, the electrode catalyst layer **42a** and the current collector **42** function as the anode, and the electrode catalyst layer **44a** and the current collector **44** function as the cathode, the gas flowing through the high pressure fluid discharging hole **38c** (see FIG. 1) is hydrogen generated by electrolysis. In this case, the mixed fluid discharged from the fluid lead-out opening **39b** (see FIG. 1) contains unreacted water that has not been electrolyzed and oxygen that has been generated by electrolysis. On the other hand, in the case where the electrolyte membrane **40** is a proton exchange membrane, the electrode catalyst layer **42a** and the current collector **42** function as the cathode, and the electrode catalyst layer **44a** and the current collector **44** function as the anode, the gas flowing through the high pressure fluid discharging hole **38c** (see FIG. 1) is oxygen that is generated by electrolysis. In this case, the mixed fluid discharged from the fluid lead-out opening **39b** (see FIG. 1) contains unreacted water that has not been electrolyzed and hydrogen generated by electrolysis.

[0031] In the case where the electrolysis cell **12** is an electrochemical hydrogen compressor cell, the electrolyte membrane **40** is a proton exchange membrane. In this case, the fluid used for electrolysis is hydrogen. The electrode catalyst layer **42a** and the current collector **42** function as the anode, and the electrode catalyst layer **44a** and the current collector **44** function as the cathode. The gas supplied to the high pressure fluid discharging hole **38c** (see FIG. 1) is hydrogen generated by electrolysis. The mixed fluid discharged from the fluid lead-out opening **39b** (see FIG. 1) contains unreacted hydrogen that has not been electrolyzed and water vapor.

[0032] The through hole **40h** is formed in the electrolyte membrane **40** at substantially the center in the radial direction. The through hole **40h** is a part of the high pressure fluid discharging hole **38c** provided in the electrolysis cell **12**. A first electrode catalyst layer **42a** is provided on a part of one surface of the electrolyte membrane **40**. The part of the one surface of the electrolyte membrane **40** on which the first electrode catalyst layer **42a** is provided is positioned between a portion surrounding a periphery of the through hole **40h** and a marginal edge portion of an outer circumferential edge. A second electrode catalyst layer **44a** is provided on a part of the other surface of the electrolyte membrane **40**. The part of the other surface of the electrolyte membrane **40** on which the second electrode catalyst layer **44a** is provided is positioned between the portion surrounding a periphery of the through hole **40h** and the marginal edge portion of the outer circumferential edge. The first electrode catalyst layer **42a** and the second electrode catalyst layer **44a** are formed respectively in, for example, a ring shape.

[0033] The electrolyte membrane **40** includes a covered portion **40a** covered with the pair of electrode catalyst layers **42a**, **44a**, and an exposed portion **40b** not covered with the electrode catalyst layers **42a**, **44a**. In the electrolysis cell **12**, a range corresponding to the covered portion **40a** in the stacking direction is an electrolysis region. The exposed portion **40b** is positioned respectively on the center side and the outer side of the electrolysis region in the radial direction, in other words, in the portion of the electrolyte membrane **40** around the through hole **40h** and the marginal edge portion of the outer circumferential edge of the electrolyte membrane **40**.

Hereinafter, of the exposed portion **40b**, the portion of the electrolyte membrane **40** around the through hole **40h** is also referred to as a first exposed portion **41a**. Further, of the exposed portion **40b**, the portion on the outer side of the electrolysis region in the radial direction (marginal edge portion of the outer circumferential edge) is also referred to as a second exposed portion **41b**.

[0034] The exposed portion **40b** of the electrolyte membrane **40** has a smaller ion exchange capacity (IEC) than the covered portion **40a**. The ion exchange capacity here is the reciprocal

(meq/g) of the weight of the electrolyte membrane **40** in a dry state required to allow exchange of 1 mol of ions.

[0035] The first electrode catalyst layer **42a** uses a ruthenium (Ru)-based catalyst, and the second electrode catalyst layer **44a** uses a platinum catalyst, for example. The first current collector **42** and the second current collector **44** are constituted, for example, from a spherical gas atomizing titanium powder sintered compact (porous conductor), for example. The first current collector **42** and the second current collector **44** are provided with a smooth surface portion on which an etching process is performed after grinding, and the porosity thereof is set within a range of 10% to 50%, and more preferably, within a range of 20% to 40%.

[0036] The inner diameter and the outer diameter of the first current collector **42** and the second current collector **44** are set so as to be positioned correspondingly to the electrolysis region. Therefore, the inner ends of the first current collector **42** and the second current collector **44** on the radial center side are arranged at interval from the high pressure fluid discharging hole **38c** in the radial direction.

[0037] The frame **42e** is fitted to the outer circumference in the first current collector **42**. The frame **42e** is configured to be denser than the first current collector **42**. The frame **42e** can be formed by extending the outer circumference of the first current collector **42** outward from the electrolysis region in the radial direction and making the extending portion dense.

[0038] The first current collector **42** is accommodated in the first chamber **45an** defined by the first separator **32**, the resin frame member **36** and the electrolyte membrane **40**. The second current collector **44** is accommodated in the second chamber **45ca** defined by the second separator **34**, the resin frame member **36** and the electrolyte membrane **40**.

[0039] A flow path member **46** is interposed between the first separator **32** and the first current collector **42** (in the first chamber **45an**), and a protective sheet member **48** is interposed between the first current collector **42** and the first electrode catalyst layer **42a**. As shown in FIG. 2, the flow path member **46** has a substantially disc-like shape, and an inlet protrusion **46a** and an outlet protrusion **46b**, which are mutually opposed in the diametrical direction, are formed on an outer circumferential portion thereof.

[0040] A supply connection path **50a**, which communicates with the fluid inlets **38a**, is formed in the inlet protrusion **46a** and is in communication with a flow path **50b** (see FIG. 3). A plurality of holes **50c** communicate with the flow path **50b** and open toward the first current collector **42**. A discharge connection path **50d**, which communicates with the flow path **50b**, is formed in the outlet protrusion **46b**, and the discharge connection path **50d** communicates with the fluid outlet **38b**.

[0041] As shown in FIGS. 2 and 3, the protective sheet member **48** has a substantially circular shape when viewed in the stacking direction, and has an inner circumference arranged inward of the inner circumference of the first current collector **42** and an outer circumference arranged at the same position as the outer circumferences of the electrolyte membrane **40** and the frame **42e**. The protective sheet member **48** is formed of an inner portion **48a** and a frame portion **48b**. The inner portion **48a** is surrounded by the frame portion **48b**. The inner portion **48a** faces the covered portion **40a**. The inner portion **48a** is arranged in the range of the electrolysis region. The outer edge of the electrolysis region and the outer edge of the inner portion **48a** coincide with each other, but are not limited thereto. A plurality of communication holes **48c** are formed in the inner portion **48a**. The frame portion **48b** is positioned radially outward of the inner portion **48a**. Rectangular holes (not shown) are, for example, formed in the frame portion **48b**.

[0042] As shown in FIG. 3, the protective sheet member **48** and the membrane electrode assembly **30** are joined at a joint **49**. The joint **49** joins the frame portion **48b** of the protective sheet member **48** to the second exposed portion **41b** of the electrolyte membrane **40**. The joint **49** is positioned between a boundary point P1 between the inner portion **48a** and the frame portion **48b** of the protective sheet member **48** and an outer circumferential end E1 of the frame portion **48b**. The joint **49** is not in contact with the boundary point P1 and is positioned away from the boundary point P1.

Similarly, the joint **49** is not in contact with the outer circumferential end **E1** and is positioned away from the outer circumferential end **E1**. Preferably, the joint **49** is provided substantially in the middle between the boundary point **P1** and the outer circumferential end **E1**.

[0043] In the present embodiment, the joint **49** is positioned outward of the boundary point **P1** and inward of a pressure-resistant member **74** to be described later. In the case where the joint **49** is positioned inwardly of the pressure-resistant member **74**, the joint **49** is not interposed between the pressure-resistant member **74** and the protective sheet member **48**. Therefore, a tolerance due to the joint **49** is less likely to be required in the thickness direction (stacking direction) of the electrolysis cell **12**.

[0044] Between the central portions of the first separator **32** and the electrolyte membrane **40**, a substantially cylindrical communication hole body **52**, which surrounds the high pressure fluid discharging hole **38c** is arranged. Hereinafter, the flow path member **46**, the first current collector **42**, and the protective sheet member **48** may be collectively referred to as a first electrode side member. In this case, the communication hole body **52** is arranged between the high pressure fluid discharging hole **38c** and the first electrode side member in the radial direction of the high pressure fluid discharging hole **38c**.

[0045] The communicating hole body **52** has an inside pipe member **54** made of a porous body facing toward the high pressure fluid discharging hole **38c**, and an outside pipe member **55** arranged between the inside pipe member **54** and the first electrode side member. Accommodating chambers **55a**, **55b** are provided on a side of the outside pipe member **55** that faces toward the inside pipe member **54**. The accommodating chambers **55a**, **55b** are formed by cutting out the radially inner side of the outside pipe member **55** into ring-like shapes at both ends in the axial direction (stacking direction), and seal members (O-rings) **56a**, **56b** are arranged in the accommodating chambers **55a**, **55b** to surround the high pressure fluid discharging hole **38c**. Owing to this feature, the high pressure fluid discharging hole **38c** is sealed from the first chamber **45an** (on the side of the first current collector **42**).

[0046] As shown in FIGS. **2** and **3**, on a side of the outside pipe member **55** facing toward the first electrode side member, a groove **55s** on which the protective sheet member **48** is disposed is formed on an end surface thereof that faces toward the electrolyte membrane **40**.

[0047] The second current collector **44** and a load applying mechanism **58** (FIG. **3**) which presses the second current collector **44** against the second electrode catalyst layer **44a** are disposed in the second chamber **45ca**. The load applying mechanism **58** is equipped with a conductive elastic member, for example, a plate spring **60**, or the like, and the plate spring **60** applies a load to the second current collector **44** via a metal plate spring holder (shim member) **62**. As the elastic member, in addition to the plate spring **60**, a disc spring, a coil spring, or the like may be used.

[0048] On the radially inner side of the electrolysis region in the second chamber **45ca**, for example, a resin sheet **68** is disposed as an insulating member that covers the first exposed portion **41a** of the electrolyte membrane **40**. The thickness of the resin sheet **68** is set to be substantially the same as the second current collector **44** and the shape of the resin sheet **68** is a ring shape with the high pressure fluid discharging hole **38c** extending substantially at the center in the radial direction. As the resin sheet **68**, for example, PEN (polyethylene naphthalate), a polyimide film, or the like may be used.

[0049] A conductive sheet **66** is disposed on the surfaces of the second current collector **44** and the resin sheet **68** on the plate spring holder **62** side. The conductive sheet **66** is constituted, for example, from a metal sheet of titanium, SUS, iron, or the like, and includes a ring-like shape with the high pressure fluid discharging hole **38c** extending at substantially the center in the radial direction.

[0050] A tubular member **70** is disposed between the load applying mechanism **58** and the high pressure fluid discharging hole **38c** in the radial direction and between the conductive sheet **66** and the second separator **34** in the stacking direction. The tubular member **70** has a cylindrical shape

made of a conductive material such as metal, for example, with the high pressure fluid discharging hole **38c** extending at substantially the center in the radial direction. At one end surface, which faces toward the second separator **34**, of the tubular member **70**, a discharge passage **71** is formed that communicates with the second chamber **45ca** and the high pressure fluid discharging hole **38c**. [0051] As described above, by arranging the communication hole body **52** (outside pipe member **55**) and the tubular member **70** between the first separator **32** and the second separator **34**, the load bearing characteristics can be improved in the vicinity of the high pressure fluid discharging hole **38c** of the electrolysis cell **12**. The electrolyte membrane **40** (first exposed portion **41a**), the resin sheet **68**, and a portion of the conductive sheet **66** on the radially inner side of the electrolysis region (a portion near the high pressure fluid discharging hole **38c**) are sandwiched between the communication hole body **52** and the tubular member **70**.

[0052] A seal member (O-ring) **72** is disposed on the radially outer side of the electrolysis region in the second chamber **45ca** so as to be interposed between the second exposed portion **41b** of the electrolyte membrane **40** and the second separator **34**. The pressure-resistant member **74** is disposed on the outer circumference of the seal member **72**. The pressure-resistant member **74** has a substantially ring-like shape, together with the outer circumferential portion thereof being fitted into the inner circumferential portion of the resin frame member **36**.

[0053] In the electrolysis cell **12**, a conductive passage electrically connected from the second separator **34** to the tubular member **70**, the conductive sheet **66**, and the second current collector **44**, and a conductive passage electrically connected from the second separator **34** to the plate spring **60**, the plate spring holder **62**, the conductive sheet **66**, and the second current collector **44** are formed.

[0054] Although not shown, a pipe connected to the high pressure fluid discharging hole **38c** is provided in the end plate **20a** shown in FIG. **1**, and a back pressure mechanism capable of regulating the discharge of gas via the high pressure fluid discharging hole **38c** is provided in the pipe.

[0055] Basically, the electrolysis stack **10** is configured as described above. Next, a method for producing the electrolysis stack **10** will be briefly described with reference to FIG. **1**.

[0056] First, on a jig (not shown) used for producing the electrolysis stack **10**, the layers are stacked in a direction opposite to the direction from top to bottom shown in FIG. **1**. That is, the end plate **20a**, the insulating plate **18a**, and the terminal plate **16a** are stacked in this order. Next, the plurality of electrolysis cells **12** are stacked on the terminal plate **16a**. Then, the terminal plate **16b**, the insulating plate **18b**, and the end plate **20b** are stacked in this order on the last stacked electrolysis cell **12**. Thereafter, the end plates **20a**, **20b** are held in an integrally fastened state by a pressing mechanism such as a plurality of tie rods **22**. In this way, the electrolysis stack **10** is produced.

[0057] Next, a method for producing the electrolysis cell **12** will be described. FIG. **4** is a flowchart illustrating a procedure of a method for producing the electrolysis cell **12**. The method for producing the electrolysis cell **12** includes a joining step **S1**, a first arrangement step **S2**, a second arrangement step **S3**, a joined body stacking step **S4**, and a multiple member stacking step **S5**.

[0058] The method for producing the electrolysis cell **12** is performed in the order of the joining step **S1**, the first arrangement step **S2**, the second arrangement step **S3**, the joined body stacking step **S4**, and the multiple member stacking step **S5**, but is not limited thereto. For example, the joining step **S1** may be performed between the first arrangement step **S2** and the second arrangement step **S3**. Alternatively, the joining step **S1** may be performed between the second arrangement step **S3** and the joined body stacking step **S4**.

[0059] The joining step **S1** is a process of forming the joint **49** by joining the frame portion **48b** of the protective sheet member **48** to a portion of the membrane electrode assembly **30** on the outer side of the covered portion **40a** (the second exposed portion **41b** of the electrolyte membrane **40**). A joined body **90** is obtained by the joining step **S1** (see FIGS. **5** and **6**). The joined body **90** is

formed of the protective sheet member **48** and the membrane electrode assembly **30** joined at the joint **49**.

[0060] As shown in FIG. 5, the joint **49** may be provided in a ring shape surrounding the inner portion **48a**. In this case, the joint **49** is provided in a ring shape concentric with the boundary position P1 on the outer side of the boundary position P1 that is formed in a substantially circular shape when viewed from the stacking direction. Further, the joint **49** is provided in the ring shape concentric with the outer circumferential end E1 on the inner side of the outer circumferential end E1 of the frame portion **48b** that is formed in a substantially circular shape when viewed from the stacking direction. The joint **49** may be provided in the ring shape concentric with the boundary position P1 and the outer circumferential end E1 in the middle of the boundary position P1 and the outer circumferential end E1 that are provided in a substantially circular shape.

[0061] Alternatively, as shown in FIG. 6, the joint **49** is concentric with the boundary position P1 or the outer circumferential end E1 formed in a substantially circular shape when viewed from the stacking direction, but may be provided in plurality at intervals from each other in the circumferential direction of the protective sheet member **48**.

[0062] The joint **49** is formed by fusion or welding, but is not limited thereto. For example, the joint **49** may be formed of an adhesive. When the joint **49** is provided in a ring shape surrounding the inner portion **48a**, the joint **49** is preferably formed by fusion or welding.

[0063] The first arrangement step S2 is a process of arranging the resin frame member **36**. To be more specific, as shown in FIG. 7, a guide rod **102** of the jig **100** is inserted into the high pressure fluid discharging hole **38h** of the resin frame member **36**, whereby the resin frame member **36** is positioned. The guide rod **102** is fixed to the placement table **104** of the jig **100** and extends along the vertical direction. Therefore, the guide rod **102** restricts the positional deviation of the resin frame member **36** in the direction orthogonal to the guide rod **102** (the radial direction of the electrolysis cells **12**).

[0064] FIG. 7 shows a state where the electrolysis cell **12** to be stacked first is produced. In this case, the resin frame member **36** is disposed on the terminal plate **16a**. Although not shown in the drawings, in a state where the n-th electrolysis cell **12** is to be stacked (n is an integer of 2 or more), the n-th resin frame member **36** is disposed on the already-stacked first separator **32** (or the second separator **34**) of the n-th electrolysis cell **12**.

[0065] The second arrangement step S3 is a step of arranging the flow path member **46** (FIG. 2), the first current collector **42** (FIG. 2), the seal members **56a**, **56b** (FIG. 2), the outside pipe member **55** (FIG. 2), and the inside pipe member **54** (FIG. 2). Specifically, as shown in FIG. 2, a guide rod **102** (FIG. 7) is inserted into the inner space of the inside pipe member **54**. Next, the seal member **56b**, the outside pipe member **55**, and the seal member **56a** are arranged around the inside pipe member **54** such that the inner pipe member **54** is inserted into the seal member **56b**, the outside pipe member **55**, and the seal member **56a**. Next, the outside pipe member **55** is inserted into the through hole **46h** of the flow path member **46**, and thus the flow path member **46** is arranged in the accommodation space **36AR** inside the resin frame member **36**. Next, the outside pipe member **55** is inserted into the through hole **42h** of the first current collector **42**, and thus the first current collector **42** is arranged in the accommodation space **36AR**. The inside pipe member **54** into which the guide rod **102** is inserted and the outside pipe member **55** into which the inside pipe member **54** is inserted to restrict the positional deviation of the flow path member **46** and the first current collector **42** in the direction orthogonal to the guide rod **102**. The inner space of the inside pipe member **54** forms a part of the high pressure fluid discharging hole **38c**.

[0066] The joined body stacking step S4 is a step of stacking the joined body **90** obtained in the joining step S1. The guide rod **102** (FIG. 7) is inserted into the through hole **40h** (FIG. 5 or 6) formed in the electrolyte membrane **40** of the joined body **90**. Thus, the joined body **90** is stacked on the first current collector **42** in a state where the protective sheet member **48** faces the first current collector **42** (FIG. 2). FIG. 8 shows a state in which the joined body **90** shown in FIG. 5 is

stacked.

[0067] The guide rod **102** is inserted into the through hole **40h** of the electrolyte membrane **40** of the joined body **90**. Therefore, the membrane electrode assembly **30** (FIG. 5) is prevented from being displaced in a direction orthogonal to the guide rod **102**. The protective sheet member **48** (FIG. 5) has been joined to the membrane electrode assembly **30**. Therefore, the protective sheet member **48** is also restricted from being displaced in the direction orthogonal to the guide rod **102**. [0068] The multiple member stacking step S5 is a step of stacking multiple of members on the joined body **90**. Specifically, the pressure-resistant member **74** (FIG. 2), the second current collector **44** (FIG. 2), the resin sheet **68** (FIG. 2), the conductive sheet **66** (FIG. 2), the plate spring holder **62** (FIG. 2), the plate spring **60** (FIG. 2), the tubular member **70** (FIG. 2), and the second separator **34** (FIG. 2) are stacked in this order. The guide rod **102** (FIG. 7) is inserted through each of these members.

[0069] When the multiple member stacking step S5 is completed, one electrolysis cell **12** is manufactured. The guide rod **102** is removed after a plurality of electrolysis cells **12** are produced (stacked) by repeating the above-described producing method. Thereafter, as described above, the terminal plate **16b**, the insulating plate **18b**, and the end plate **20b** are stacked in this order, and the end plates **20a**, **20b** are integrally fastened.

[0070] In a case where the protective sheet member **48** is not joined to the membrane electrode assembly **30** at the joint **49**, a restricting member is required to restrict positional displacement of the protective sheet member **48** in a direction orthogonal to the guide rod **102** (radial direction of the electrolysis cell **12**). The reason therefor is as follows. That is, as shown in FIG. 2, in order to dispose the seal member **56a** (FIG. 2), the through hole **48h** (FIG. 2) of the protective sheet member **48** should be larger than the through hole **40h** of the electrolyte membrane **40**. Therefore, in a stage before the end plates **20a**, **20b** are integrally fastened, there is a concern that the relatively large clearance existing between the through hole **48h** (FIG. 2) of the protective sheet member **48** and the guide rod **102** inserted into the through hole **48h** may cause positional deviation of the protective sheet member **48**. Therefore, when the protective sheet member **48** is not joined to the membrane electrode assembly **30** at the joint **49**, another restricting member is required.

[0071] In contrast, in the present embodiment, the protective sheet member **48** is stacked on the fluid-supply-side current collector (first current collector **42**) after being joined to the membrane electrode assembly **30** at the joint **49**. The clearance between the through hole **40h** of the electrolyte membrane **40** of the membrane electrode assembly **30** and the guide rod **102** inserted into the through hole **40h** is small. Thus, the guide rod **102** restricts displacement of the protective sheet member **48** in a direction orthogonal to the guide rod **102** via the membrane electrode assembly **30**. Therefore, even if there is no restricting member dedicated to the protective sheet member **48**, the positional deviation of the protective sheet member **48** in the direction orthogonal to the guide rod **102** can be restricted. In this way, the risk of misalignment during the manufacture of the electrolysis cell **12** can be reduced.

[0072] In the present embodiment, as shown in FIG. 3, the joint **49** is positioned away from each of the boundary position P1 between the inner portion **48a** and the frame portion **48b** of the protective sheet member **48** and the outer circumferential end E1 of the frame portion **48b**. Therefore, the joint strength between the protective sheet member **48** and the membrane electrode assembly **30** can be increased as compared to the case where the joint **49** is not away from the boundary position P1 or the outer circumferential end E1.

[0073] In the present embodiment, the joint **49** is formed by fusion or welding. Therefore, the protective sheet member **48** and the membrane electrode assembly **30** can be joined together more easily without distortion than in the case where the joint **49** is formed of an adhesive or the like.

[0074] As shown in FIG. 5, when the joint **49** is formed in a ring shape surrounding the inner portion **48a**, the joint strength between the protective sheet member **48** and the membrane electrode assembly **30** can be increased. In this case, the formation of the joint **49** by fusion or welding is

facilitated.

[0075] The following supplementary notes are further disclosed in relation to the above embodiment.

Supplementary Note 1

[0076] The method for producing the electrolysis cell (12) according to the present disclosure includes: the membrane electrode assembly (30) including the electrolyte membrane (40) with the through hole (40h) and the pair of electrode catalyst layers (42a, 44a) provided respectively on both surfaces of the electrolyte membrane and covering the electrolyte membrane in the range extending between the portion surrounding the periphery of the through hole to the marginal edge portion of the outer circumference; the pair of current collectors (42, 44) provided respectively on both sides of the membrane electrode assembly; and the protective sheet member (48) disposed between one of the pair of current collectors and the membrane electrode assembly, wherein the one of the pair of current collectors is the fluid-supply-side current collector (42) to which the fluid to be used for electrolysis is supplied, and the protective sheet member includes the inner portion (48a) facing the electrolyte membrane in the range (40a) covered with the electrode catalyst layers and the frame portion (48b) surrounding the inner portion, the method for producing the electrolysis cell including the joining step (S1) of forming the joint (49) by joining the frame portion to the portion of the membrane electrode assembly at the position outward of the range covered with the catalyst layers; and the joined body stacking step (S4) of stacking on the fluid-supply-side current collector the membrane electrode assembly and the protective sheet member that have been joined, in a manner that the protective sheet member faces the fluid-supply-side current collector.

Supplementary Note 2

[0077] In the method for producing an electrolysis cell according to Supplementary Note 1, the joint may be positioned away from each of the outer circumferential end (E1) of the frame portion and the boundary position (P1) between the inner portion and the frame portion.

Supplementary Note 3

[0078] In the method for producing an electrolysis cell according to Supplementary Note 1, the joint may be provided in plurality at intervals from each other in the circumferential direction of the protective sheet member.

Supplementary Note 4

[0079] In the method for producing an electrolysis cell according to Supplementary Note 1, the joint may be formed by fusion or welding.

Supplementary Note 5

[0080] In the method for producing an electrolysis cell according to Supplementary Note 4, the joint may be formed in the ring shape surrounding the inner portion.

[0081] Although concerning the present disclosure, a detailed description thereof has been presented above, the present disclosure is not necessarily limited to the individual embodiments described above. These embodiments may be subjected to various additions, substitutions, modifications, partial deletions and the like, within a range that does not deviate from the essence and gist of the present disclosure, or the spirit of the present disclosure as derived from the contents described in the claims and equivalents thereof. Further, the embodiments can also be implemented together in combination. For example, in the above-described embodiments, the order of each of the operations and the order of each of the processes are illustrated as examples, and the present invention is not necessarily limited to these features. The same also applies to cases in which numerical values or mathematical expressions are used in the description of the aforementioned embodiments.

Claims

1. A method for producing an electrolysis cell comprising: a membrane electrode assembly including: an electrolyte membrane with a through hole; and a pair of electrode catalyst layers provided respectively on both surfaces of the electrolyte membrane and covering the electrolyte membrane in a range extending between a portion surrounding a periphery of the through hole to a marginal edge portion of an outer circumferential edge; a pair of current collectors provided respectively on both sides of the membrane electrode assembly; and a protective sheet member disposed between one of the pair of current collectors and the membrane electrode assembly, wherein the one of the pair of current collectors is a fluid-supply-side current collector to which a fluid to be used for electrolysis is supplied, and the protective sheet member includes an inner portion facing the electrolyte membrane in the range covered with the electrode catalyst layers and a frame portion surrounding the inner portion, the method for producing the electrolysis cell comprising: a joining step of forming a joint by joining the frame portion to a portion of the membrane electrode assembly at a position outward of the range covered with the catalyst layers; and a joined body stacking step of stacking on the fluid-supply-side current collector the membrane electrode assembly and the protective sheet member that have been joined, in a manner that the protective sheet member faces the fluid-supply-side current collector.
 2. The method for producing the electrolysis cell according to claim 1, wherein the joint is positioned away from each of an outer circumferential end of the frame portion and a boundary position between the inner portion and the frame portion.
 3. The method for producing the electrolysis cell according to claim 1, wherein the joint is provided in plurality at intervals from each other in a circumferential direction of the protective sheet member.
 4. The method for producing the electrolysis cell according to claim 1, wherein the joint is formed by fusion or welding.
 5. The method for producing the electrolysis cell according to claim 4, wherein the joint is formed in a ring shape surrounding the inner portion.
-